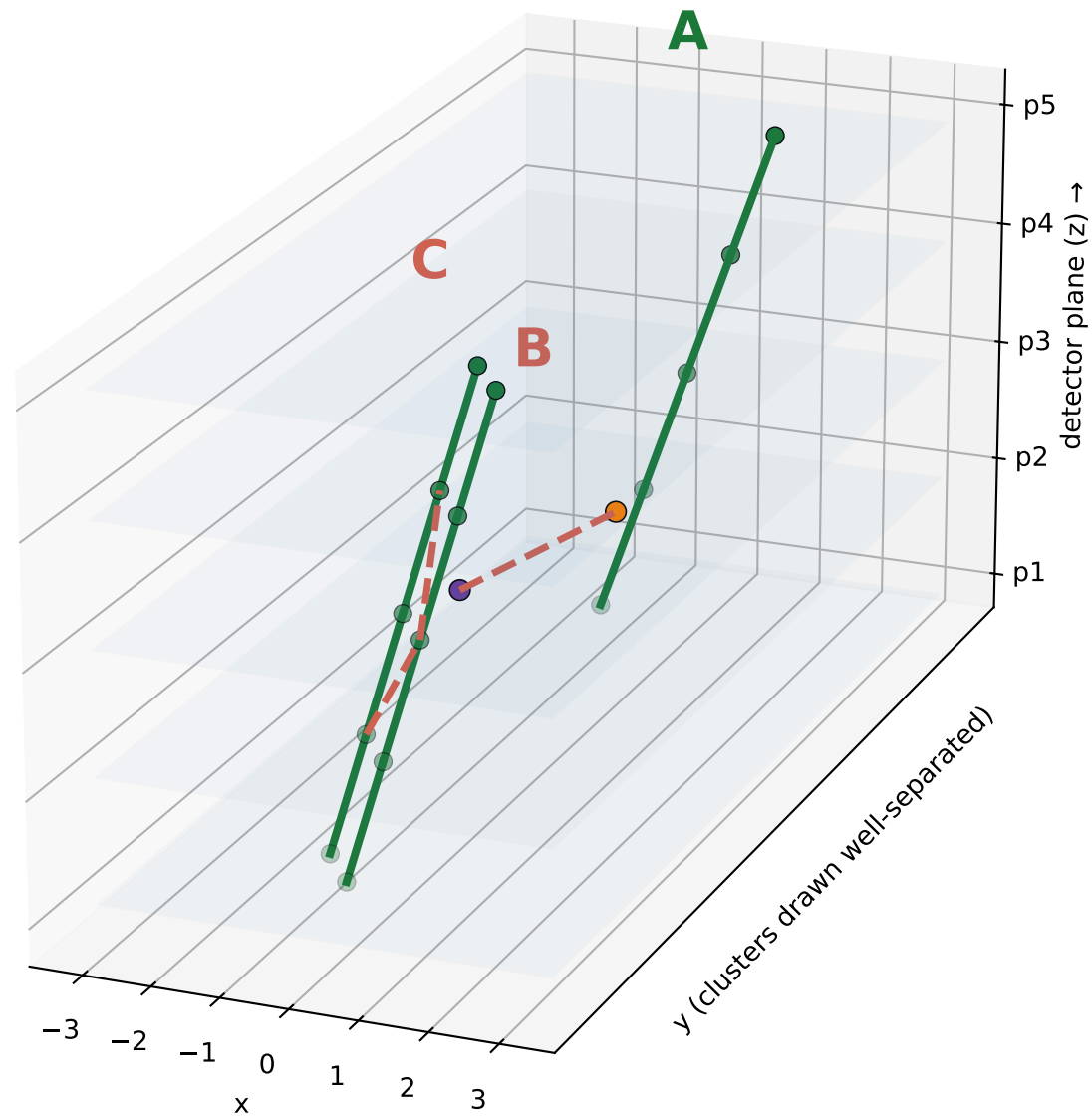
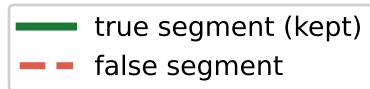


# Schematic: the three segment types and what the 1-bit quantum filter does to each



A real track (4-segment chain): eigenvalues  $\{2.4, 3.4, 4.6, 5.6\}$  are OFF the notch  $\rightarrow$  quantum KEEPS it (correct; keeps 3/4)  
B isolated false segment (no compatible neighbour): eigenvalue =  $\gamma + \delta = 4 =$  the NOTCH  $\rightarrow$  quantum ERASES it (correct)  
C cross-track bridge (two near-parallel tracks accidentally aligned): eigenvalues OFF the notch  $\rightarrow$  quantum KEEPS it (FALSE POSITIVE)